Technical reference manual
1.0.0

EtherBench

Pinout & Hardware Mapping

The core of the EtherBench motherboard features an STM32H5 in a VFBGA201 package, exposing an extensive array of hardware peripherals. To streamline firmware development (BSP handle definition) and facilitate non-intrusive laboratory troubleshooting, all active pins are detailed below.

Testpoints (TP) are specified only for signals where trace velocity, parasitic capacitance, and hardware layout constraints allow safe physical probing.

Host communication

These signals are used as the core functions, such as host communication or flash storage.

Instance	Signal	Pin	BGA	Description
`USBHS1`	`DP` `DM`	`PA12` `PA11`	`B15` `C15`	USB peripheral, routed to the hub as a device.
`USB_SUSP`	`USB_SUSP`	`PD2`	`D12`	USB Upstream suspended signal.
`USB_HS`	`USB_HS`	`PD3`	`D11`	USB Upstream as high-speed signal.
`USB_RESET`	`USB_RESET`	`PH13`	`E12`	USB hub reset signal

`UCPD1`	`CC1`
	`CC2`
	`DB_CC1`
	`DB_CC2`
		`PB13`
		`PB14`
		`PA9`
		`PC9`
			`P13`
			`R14`
			`E15`
			`F14`
				USB-C signals. Routed directly to the USB-C port.

`ETH`	`CLK`
	`MDC`
	`MDIO`
	`CRS_DV`
	`TX_EN`
	`RXD0`
	`RXD1`
	`TXD0`
	`TXD1`
		`PA1`
		`PC1`
		`PA2`
		`PA7`
		`PB11`
		`PC4`
		`PC5`
		`PB12`
		`PB15`
			`N2`
			`M3`
			`P2`
			`R3`
			`R13`
			`N5`
			`P5`
			`P12`
			`R15`
				Ethernet signals, routed to the PHY.

`ETH_RESET`	`ETH_RESET`	`PD9`	`P14`	Ethernet PHY reset

`OCTOSPI1`	`CLK`
	`IO0`
	`IO1`
	`IO2`			

	IO3 NCS	PF10 PF8 PF9 PF7 PF6 PE11	L1 L3 L2 K1 K2 P10	QSPI Flash signals.
DEBUG	SWCLK SWDIO	PA14 PA13	A14 A15	SWD signals, routed to a standard header for initial flash.
SDMMC2	SCK CMD D0 D1 D2 D3	PD6 PD7 PG9 PG10 PG11 PG12	B11 A11 C10 B10 B9 B8	SD systems. Limited to 3.3V bus.
EVENTOUT	TRIG_OUT	PD10	N15	Trigger output.
EXTI	TRIG_IN	PD8	N15	Trigger input

Programmer

These signals are the one used as the SWD/JTAG signals to flash the probe. They're placed appart, as they're runned not used as standard peripherals.

Instance	Signal	Pin	BGA	Description
SWD / JTAG	SCLK MISO MOSI IO	PC10 - PI1 PI2 PI3 PC12	D14 - B14 C14 C13 A12	DUT Programmation signals.

Operation principle

As the SWD bus require a bi-directionnal communication, we used an SPI peripheral that can do both, here, SPI3. The SPI2 peripheral is used for JTAG, where TDI and TDO pins are used. The second SPI slave run as a full-duplex slave to ensure synchronisation of the two peripherals.

When running in SWD mode, the SPI3 peripheral can be reconfigured to a serial receiver to provide the SWO ability.

DUT communication

These signals are used by the hardware IO to the DUT, on the standard serial interfaces.

Instance	Signal	Pin	BGA	Description
`MCO1`	`MCO1`	`PA8`	`F15`	Clock output for the DUT.
`I3C1`	`SCL` `SDA`	`PB8` `PB9`	`A5` `B4`	I3C master / slave.

`USART10`	`TX`
	`RX`
	`RTS`
	`CTS`
		`PE3`
		`PE2`
		`PG14`
		`PG13`
			`A1`
			`A2`
			`A7`
			`A8`
				Full featured USART bus.

`USART5`	`TX` `RX`	`PB6` `PB5`	`B6` `A6`	Rx/Tx only featured USART bus.
`USART11`	`TX` `RX`	`PF3` `PF4`	`J2` `J3`	Rx/Tx only featured USART bus.

`SPI4`	`SCLK`
	`MISO`
	`MOSI`
	`NSS0`
	`NSS1`
		`PE12`
		`PE5`
		`PE6`
		`PI0`
		`PC7`
			`R10`
			`B2`
			`B3`
			`E14`
			`G15`
				Full duplex SPI Master. NSS are done by GPIOs.

`FDCAN1`	`TX` `RX`	`PE1` `PE0`	`A3` `A4`	FDCAN peripheral.
`ADC1`	`IN2` `IN5`	`PF11` `PB1`	`R6` `R4`	Analog input.
`DAC1`	`OUT1` `OUT2`	`PA4` `PA5`	`N4` `P4`	Analog output.

`GPIO`	`OD_W_1`
	`OD_W_2`
	`OD_R_1`
	`OD_R_2`
	`PP_1`
	`PP_2`
		`PI8`
		`PI9`
		`PI11`
		`PH2`
		`PI5`
		`PI4`
			`D2`
			`D3`
			`D4`
			`C4`
			`E4`
			`F4`
				GPIO (2 pair of open drain OUT + READ + 2 push pull).

`BUCK_EN`	`EN`	`PI7`	`C2`	DUT buck enable
`BUCK_INT`	`INT`	`PI6`	`C3`	DUT buck inerrupt

DUT_RESET	RESET	PH5	J4	DUT reset signal
DUT_PRESENT	PRESENT	PH4	H4	DUT present signal

User IO

These signals are used by the hardware to show informations to the user.

Instance	Signal	Pin	BGA	Description
TIM13	aRGB_OUT	PA6	P3	aRGB output for the led ring.
TIM1 - TIM2	User leds	PE9 - PE13 - PE14 - PH9 - PA3 - PA15 - PB3 - PB10	P9 - N11 - P11 - M13 - R2 - A13 - A10 - R12	PWM Leds outputs.
GPIO_INPUT	User buttons	PH6 - PH7 - PH8	M11 - M12 - N12	User buttons
GPIO_INPUT	User reset	PD12	N13	User reset request.

Miscellaneous

These signals describe all the others pins, for a lot of different functions that may not be linked together.

Instance	Signal	Pin	BGA	Description
I2C2	SCL SDA	PF1 PF0	H3 E2	I2C for management and internal usage.
I2C4	SCL SDA ALERT	PH11 PH12 PH10	L12 K12 L13	SMBUS for the USB hub management.
USART9	TX RX IN OUT	PD15 PD14 PD13 PG2	L14 M14 M15 L15	Serial bus for the INTERCOM system.
TIM3	PWM1	PC6	H15	PWM Output for the optionnal fan.

`RCC_8M`	`IN` `OUT`	`PH0` `PH1`	`G1` `H1`	Crystal for the 8 MHz clock source.
`RCC_32kHz`	`IN` `OUT`	`PC14` `PC15`	`E1` `F1`	Crystal for the 32.768 kHz Clock source.

`DEBUG`	`DBG_01`
	`DBG_02`
	`DBG_03`
	`DBG_04`
	`DBG_IN1`
	`DBG_IN2`
	`DBG_LED1`
	`DBG_LED2`
		`PG1`
		`PG0`
		`PF15`
		`PF14`
		`PG7`
		`PG6`
		`PG4`
		`PG3`
			`N7`
			`M7`
			`P7`
			`R7`
			`J14`
			`J15`
			`K14`
			`K15`
				Custom debug pins for the firmware build.

`PRESENT`	`BOARD_PRESENT`	`PC13`	`D1`	Daughter board presence detection.
`SCREWED`	`SCREWED`	`PE4`	`B1`	Daughter board screwed detection.
`OE`	`OE`	`PF2`	`H2`	Buffer output enable.
`POWER_OK`	`PGOOD`	`PE7`	`R8`	Master regulator power good signal.